

Differential 1-forms

– Dual Object of a Vector Field –

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수학의 즐거움, *Enjoying Math*

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Contents



I. What is a Vector Field?

II. Cotangent Space

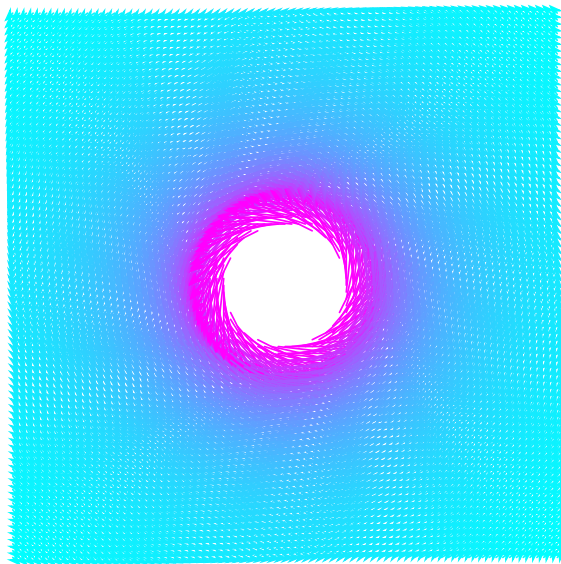
III. Conclusion

I. What is a Vector Field?



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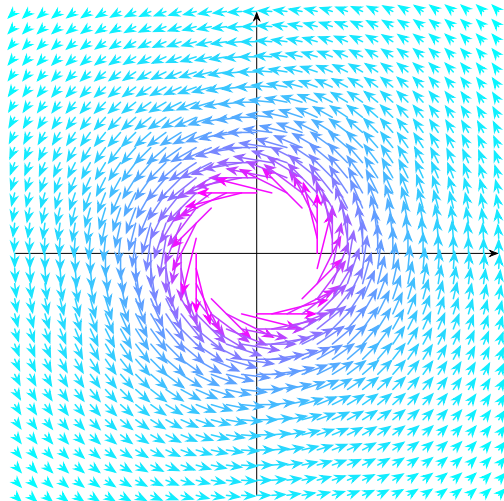
Example





I. What is a Vector Field?

Example



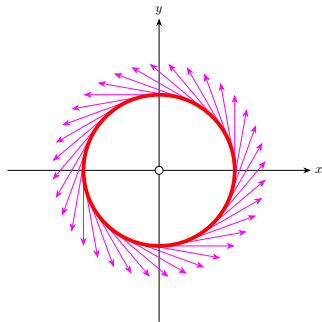
Consider

$$\mathbf{F}(x, y) = \begin{bmatrix} \frac{-y}{x^2 + y^2} \\ \frac{x}{x^2 + y^2} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} \theta \\ \frac{\partial}{\partial y} \theta \end{bmatrix}$$

with $\theta = \theta(x, y) = \arctan(y/x)$.

I. What is a Vector Field?

Tangent space $T_p S^1$ (at a point $p \in S^1$)



Let

$$S^1 := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

and consider the vector field

$$\mathbf{F}(x, y) = \begin{bmatrix} -y \\ x \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Then

$$T_p S^1 = \text{span} \left(\begin{bmatrix} -y \\ x \end{bmatrix} \right).$$

Note. Fix $p \in S^1$. $T_p S^1$ is a one-dimensional line through the origin, spanned by $\mathbf{F}(p)$.



I. What is a Vector Field?

Definition of a Vector Field with Tangent Space

Define

$$TS^1 := \bigsqcup_{p \in S^1} T_p S^1 = \left\{ (p, \mathbf{v}) : p \in S^1, \mathbf{v} \in T_p S^1 \right\}.$$

A **vector field** (on S^1) is a map

$$\begin{aligned} \mathbf{X} : S^1 &\longrightarrow TS^1 \\ p &\longmapsto (p, \mathbf{X}(p)) \end{aligned},$$

where $\mathbf{X}(p) \in T_p S^1$.

II. Cotangent Space

II. Cotangent Space

Vectors and Covectors



“Vectors = columns” and “Covectors = rows”

II. Cotangent Space

Vectors and Covectors



Fix any point $p \in S^1$.

- A vector $\mathbf{v} \in T_p S^1$ is a column

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

satisfying $xv_1 + yv_2 = 0$.

- A covector λ is a linear functional on $T_p S^1$; represent it as a row **row** $[\alpha \quad \beta]$ acting by

$$\lambda(\mathbf{v}) = [\alpha \quad \beta] = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \alpha v_1 + \beta v_2.$$

II. Cotangent Space

Cotangent Space



Define

$$T_p^*S^1 := \text{Hom}(T_pS^1, \mathbb{R}) = \left\{ \text{rows acting linearly on columns in } T_pS^1 \right\}.$$

Note. Columns live in T_pS^1 , rows live in $T_p^*S^1$.



II. Cotangent Space

1-form as a dual of vector field

Define

$$TS^1 := \bigsqcup_{p \in S^1} T_p S^1 = \left\{ (p, \mathbf{v}) : p \in S^1, \mathbf{v} \in T_p S^1 \right\}$$

$$T^* S^1 := \bigsqcup_{p \in S^1} T_p^* S^1 = \left\{ (p, \lambda) : p \in S^1, \lambda \in T_p^* S^1 \right\}.$$

Note.

- A **vector field** is a map $\mathbf{X} : S^1 \rightarrow TS^1$.
- A **1-form** is a map $\omega : S^1 \rightarrow T^* S^1$.

For example,

$$\mathbf{F}(x, y) = \begin{bmatrix} -y \\ x \end{bmatrix}, \quad \omega = (-y)dx + xdy.$$

II. Cotangent Space

Line integrals: vector vs. 1-form



Let $\gamma : [a, b] \rightarrow S^1$.

- **Line Integral of Vector-field:**

$$\int_{S^1} \mathbf{F} \cdot d\mathbf{r} = \int_{[a,b]} \mathbf{F}(\gamma(t)) \cdot \gamma'(t) dt.$$

- **Line Integral of 1-form:**

$$\int_{S^1} \omega = \int_{[a,b]} \omega_{\gamma(t)}(\gamma'(t)) dt.$$

III. Conclusion



Vectors live in TM ; 1-forms live in T^*M and pair with vectors (duality).

The End

Questions? Comments?